Worksheet 1-4 A: Limits From Tables #1
AP Calculus

Name _____

Use the table of values to determine the limit.

1.) $\lim_{x \to 9} f(x) =$ _____

x	8.7	8.999	9.001	9.8
$f(x)$	−5.8	−5.001	−4.999	−4

2.) $\lim_{x \to -7} g(x) =$ _____

x	−7.5	−7.001	−6.999	−6.5
$g(x)$	3.8	3.501	3.499	3.2

3.) $\lim_{x \to -2} h(x) =$ _____

x	−2.1	−2.001	−1.999	−1.9
$h(x)$	−8.7	−8.999	9.001	9.4

4.) $\lim_{x \to 11} m(x) =$ _____

x	10.7	10.99	11.01	11.3
$g(x)$	10.3	10.001	9.999	9.6

Evaluate each limit by creating a table of values. Remember to always round (or truncate) answers to three decimal places.

5.) $\lim_{x \to -5} \frac{x^2-2x-35}{x+5} =$ _____

x						
$f(x)$						

6.) $\lim_{x \to -2} \frac{x^2+5x+6}{x+2} =$ _____

x						
$f(x)$						

7.) $\lim_{x \to 2} \frac{x^2+4x-12}{x-2} =$ _____

x						
$f(x)$						

8.) $\lim_{x \to 1} \frac{5x^3+2x^2-13x+6}{x-1} =$ _____

x						
$f(x)$						

Use the information given for each problem to evaluate the limit. Remember to always round (or truncate) answers to three decimal places.

9.) The function f is continuous and decreasing for $x \geq -5$. The table gives values of f at selected values of x.

x	−3.1	−3.01	−2.99	−2.8
$f(x)$	−3.4	−3.499	−3.501	−3.8

Approximate the value of:

$\lim_{x \to -3} \ln(-f(x)) =$ _____

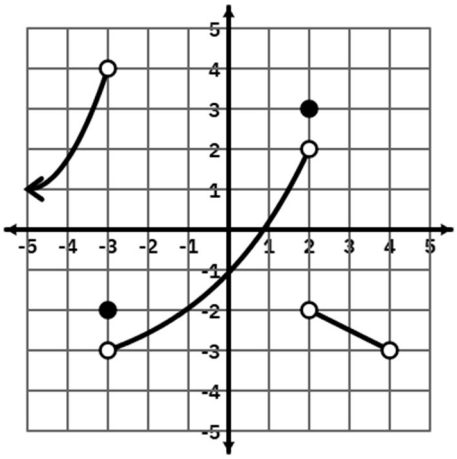
10.) The function f is continuous and increasing for $x \geq -7$. The table gives values of f at selected values of x.

x	−5.1	−5.001	−4.999	−4.8
$f(x)$	3.7	3.999	4.001	4.2

Approximate the value of:

$\lim_{x \to -5} \sqrt[5]{f(x)} =$ _____

11.) Give the value of each statement

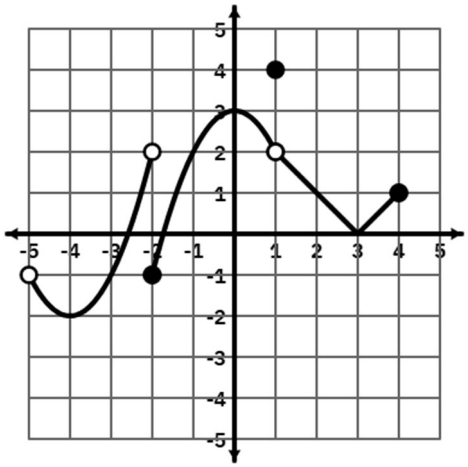


a.) $\lim_{x\to 2} f(x) =$ _____ b.) $f(-3) =$ _____ c.) $\lim_{x\to -3^-} f(x) =$ _____

d.) $\lim_{x\to 2^+} f(x) =$ _____ e.) $f(2) =$ _____ f.) $\lim_{x\to 2^-} f(x) =$ _____

g.) $\lim_{x\to -3^+} f(x) =$ _____ h.) $f(4) =$ _____ i.) $\lim_{x\to -3} f(x) =$ _____

12.) Give the value of each statement



a.) $\lim_{x\to 1} f(x) =$ _____ b.) $f(-2) =$ _____ c.) $\lim_{x\to -2^+} f(x) =$ _____

d.) $\lim_{x\to 2} f(x) =$ _____ e.) $f(-4) =$ _____ f.) $\lim_{x\to 1^-} f(x) =$ _____

g.) $\lim_{x\to 1^+} f(x) =$ _____ h.) $f(-5) =$ _____ i.) $f(1) =$ _____

13.) The table below shows values of the function f at selected values of x. Which of the following is true based on the data from the table?

x	8.9	8.99	8.999	9.001	9.01	9.1
$f(x)$	0.7	0.8	0.999	2.001	2.01	2.3

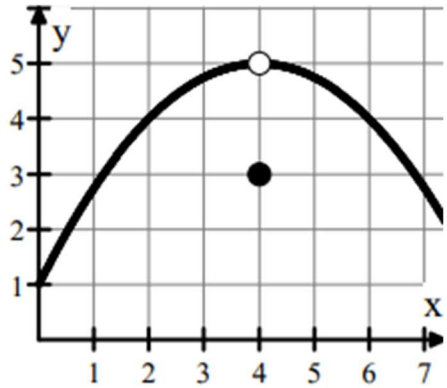
(A) $\lim_{x\to 9} f(x) = 1$

(B) $\lim_{x\to 9} f(x) = 2$

(C) $\lim_{x\to 9^-} f(x) = 2$ and $\lim_{x\to 9^+} f(x) = 1$

(D) $\lim_{x\to 9^-} f(x) = 1$ and $\lim_{x\to 9^+} f(x) = 2$

14.) The graph of the function f is shown at the right. The value of $\lim_{x\to 4} 2\cos(f(x))$ is



(A) 0.567
(B) –1.307
(C) –1.979
(D) Does not exist